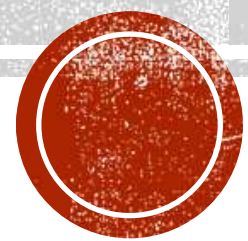
GFN1000
IN DIALOGUE WITH NATURE
与自然对话



Newton & Cohen

Aristotle's Philosophy of Nature

Metaphysics & Epistemology

Nature and Change
– how change happens and be explained

Cosmology
– Aristotelian universe: sublunar and supralunar region

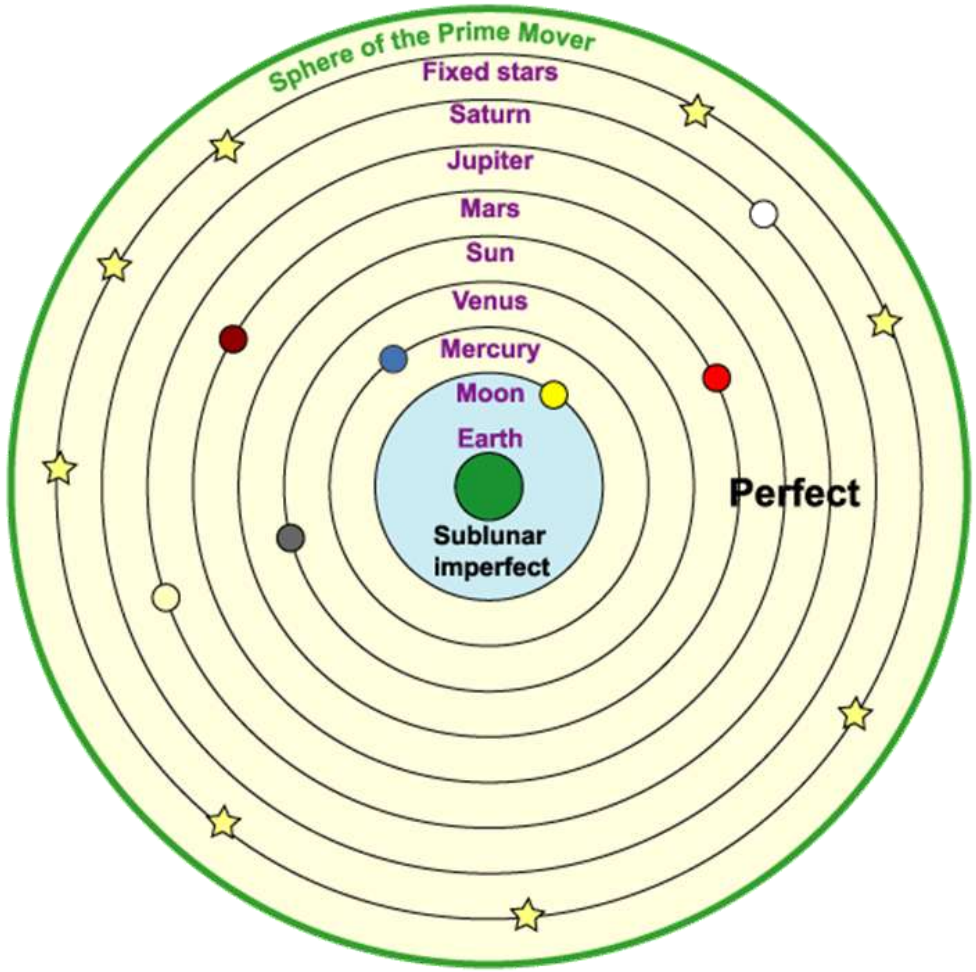
Motion
– terrestrial and celestial motions



Aristotle (384-322BC)



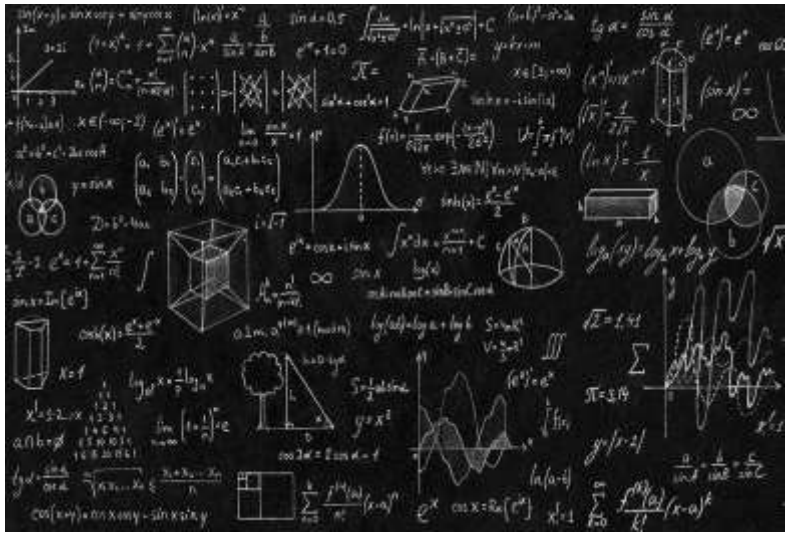
- **A geocentric (地心的) universe**



Core Question

- What is scientific understanding?



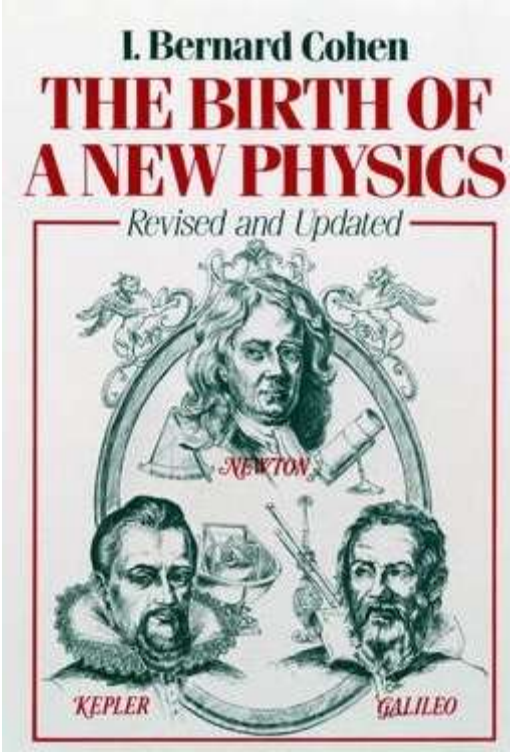


Content

Text 3a: I. Bernard Cohen

- "*The Birth of a New Physics*":

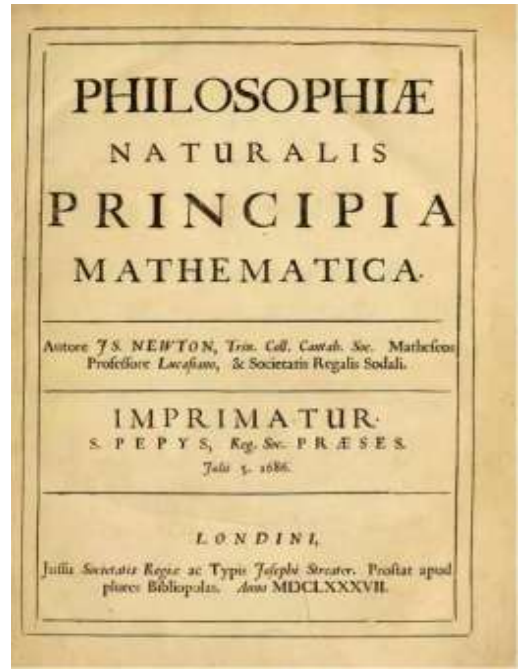
  Chapter 7 The Grand Design-a New Physics



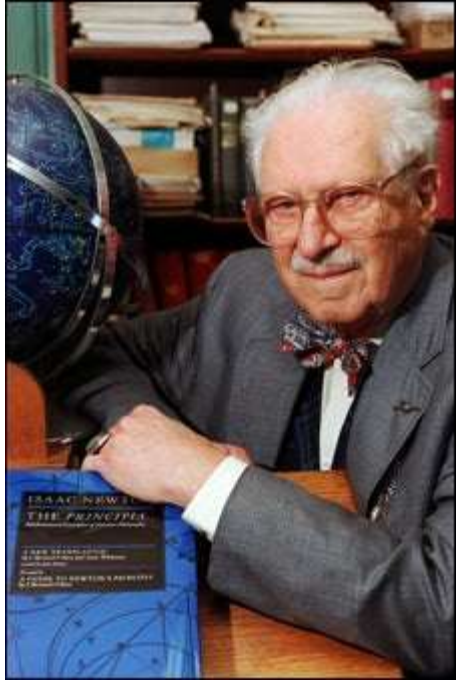
Text 3b: Isaac Newton

- "*The Principia*":

  Definitions



I. Bernard Cohen (1914-2003)



- First American to receive a PhD in history of science
- Victor S. Thomas Professor of the history of science at Harvard University
- Internationally recognized Newton scholar
- Recipient of the Sarton medal *, in 1974.

*The most prestigious award of the History of Science Society

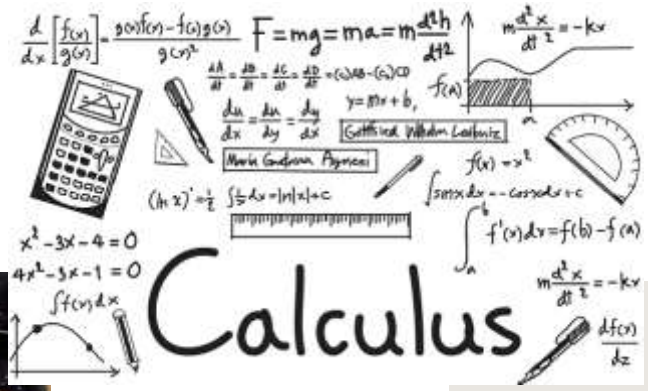
Isaac Newton – A Short Biography

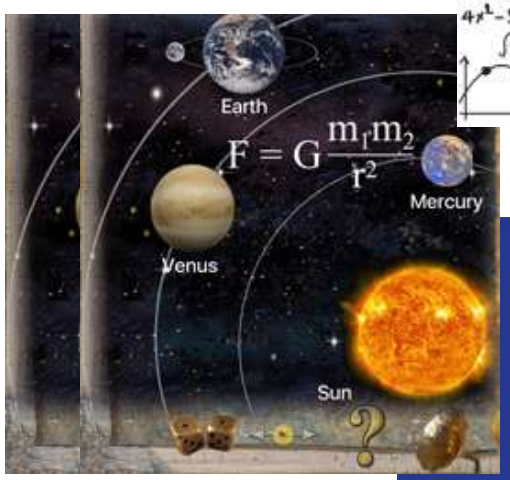
- 1643 Born to a family of yeoman farmers;
- 1661 Entered Trinity College of Cambridge University;
- 1665-1667 University closed due to the plague (瘟疫); Newton made most significant discoveries (optics, calculus, gravity) during this time;
- 1669-1702 *Lucasian Professor of Mathematics* at the University of Cambridge;
- 1687 Published *Principia;* (three laws of motion)
- 1699 Master of Royal Mint 皇家铸币厂;
- 1703-1727 President of the Royal Society;
- 1727 Died in London



Sir Isaac Newton (1642-1726)

- Mathematician
- Astronomer
- Physicist
- President of the Royal Society









Described in his own day as a "natural philosopher"

Sir Isaac Newton (1642-1726)



Mathematics

- Differential and integral calculus (微积分)(Leibniz/莱布尼茨)

  Newton's notation

  $\dot{f} \quad \ddot{f}$

  Leibniz's notation

  $\frac{df}{dt} \quad \frac{d^2f}{dt^2}$

- Binomial theorem (二项式定理)

$$(x+y)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^{\alpha-k} y^k \text{。其中} \binom{\alpha}{k} = \frac{\alpha(\alpha-1)\dots(\alpha-k+1)}{k!} = \frac{(\alpha)_k}{k!}\text{。}$$

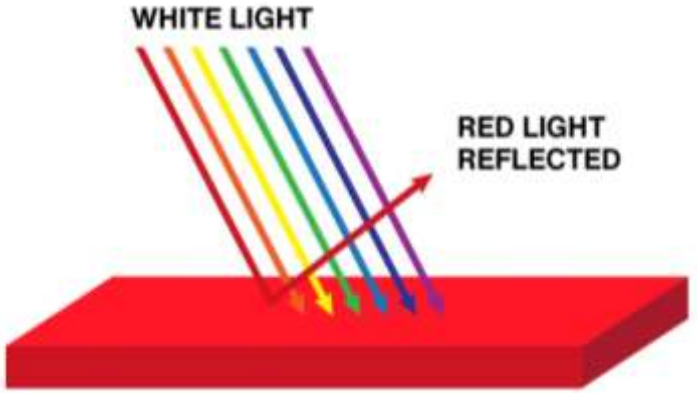
- Newton-Raphson method (quickly find a good approximation for the root of a real-valued function)

Sir Isaac Newton (1642-1726)

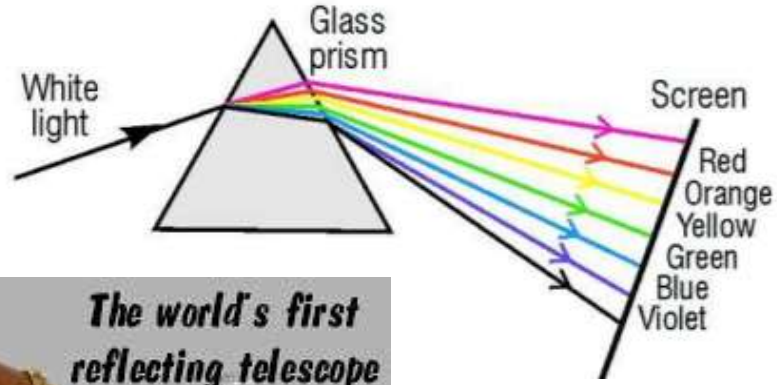


Optics

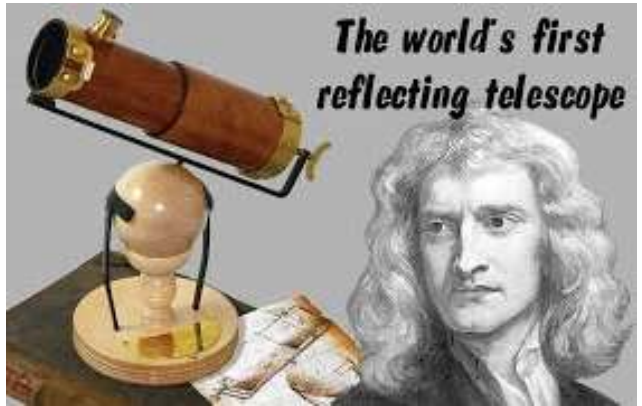
- Color is the result of objects interacting with already-colored light rather than objects generating the color themselves.



- White light is a mixture of light of many colors, each having a characteristic index of refraction (折射).



- Invented a reflecting telescope (反射望远镜) and showed astronomers how to transcend (超越) the limitations of telescopes built of lenses (透镜)



Sir Isaac Newton (1642-1726)



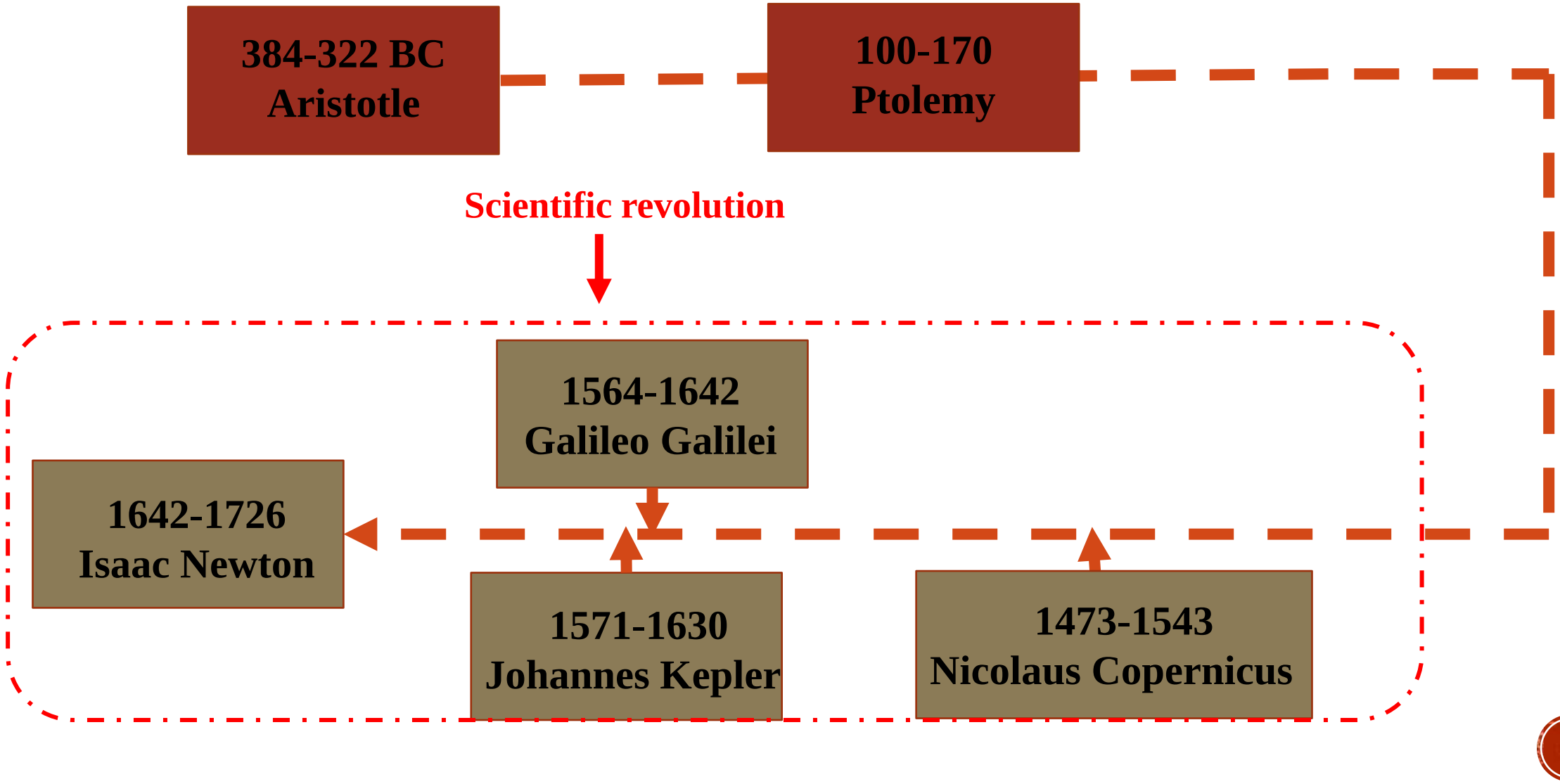
Celestial mechanics

- What makes an apple falls down is responsible for keeping the moon in orbit.

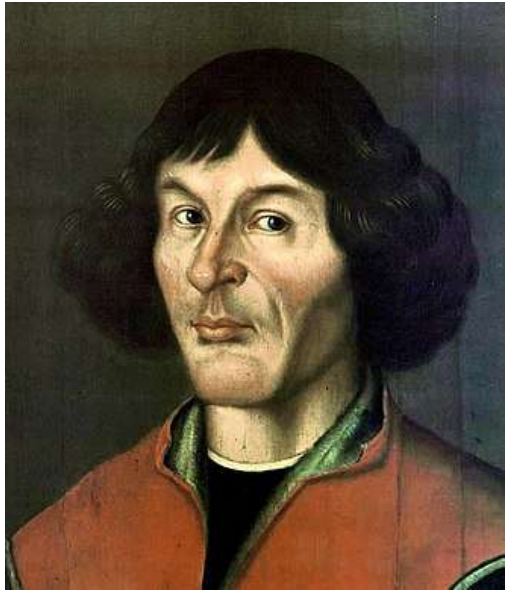
- He unified celestial (天体的) and terrestrial (地面的) dynamics.

$$F = G\frac{m_1 m_2}{r^2}$$

F is the gravitational force acting between two objects, m_1 and m_2 are the masses of the objects, r is the distance between the centers of their masses, G is the gravitational constant.

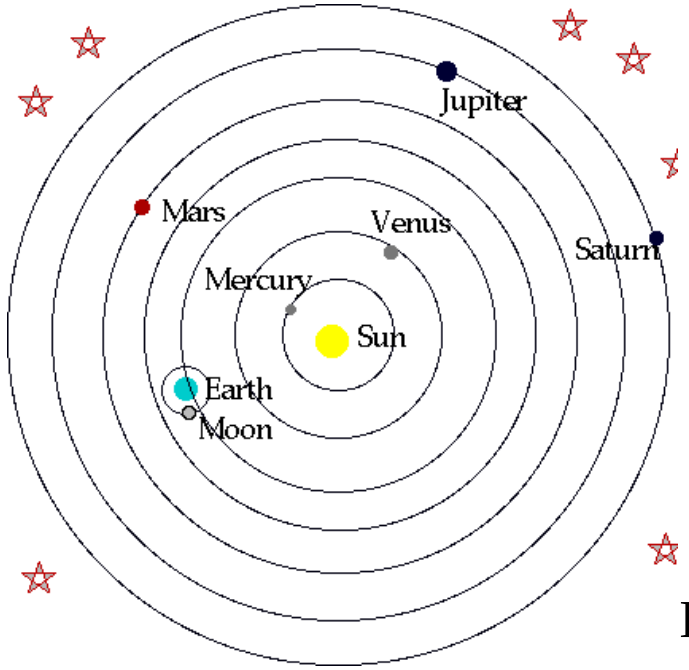

384-322 BC
Aristotle
100-170
Ptolemy
Scientific revolution
1564-1642
Galileo Galilei
1642-1726
Isaac Newton
1571-1630
Johannes Kepler
1473-1543
Nicolaus Copernicus

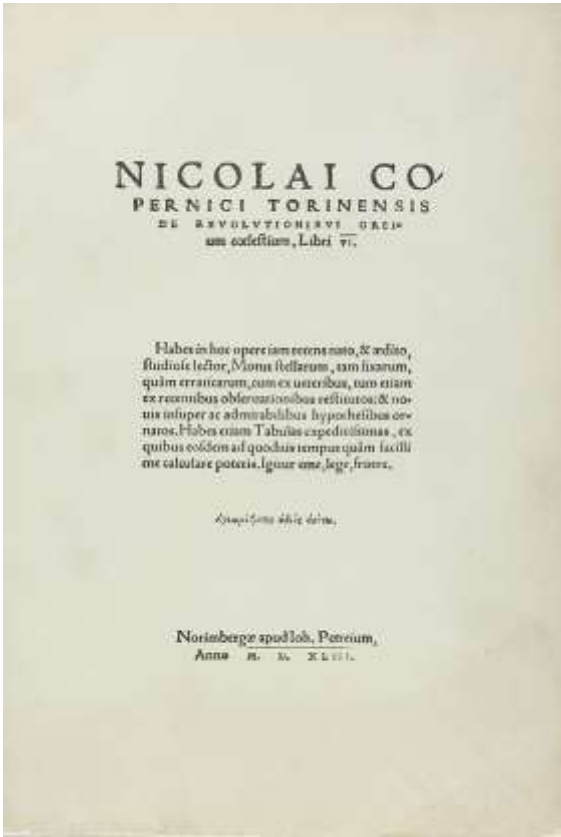
Nicolas Copernicus (1473-1543)



- Heliocentric theory (日心说)

The sun is the center of the universe and the Earth and other planets revolve around it.




NICOLAI CO
PERNICI TORINENSIS
DE REVOLVTIONIBVS ORBI
um coelestium, Libri VI.

Habes in hoc opere iam recens nato, & ædito, studiose lector, Motus stellarum, tam fixarum, quàm errantium, cum ex ueteribus, tum etiam ex recentibus obseruationibus restitutos: & nouis insuper ac admirabilibus hypothesibus ornatos. Habes etiam Tabulas expeditissimas, ex quibus eosdem ad quoduis tempus quàm facillime calculare poteris. Igitur eme, lege, fruere.

Norimbergæ apud Ioh. Petreium,
Anno M. D. XLIII.

《On the Revolutions of the Heavenly Spheres》
De revolutionibus orbium coelestium
《天体运行论》

Galileo Galilei (1564-1642)



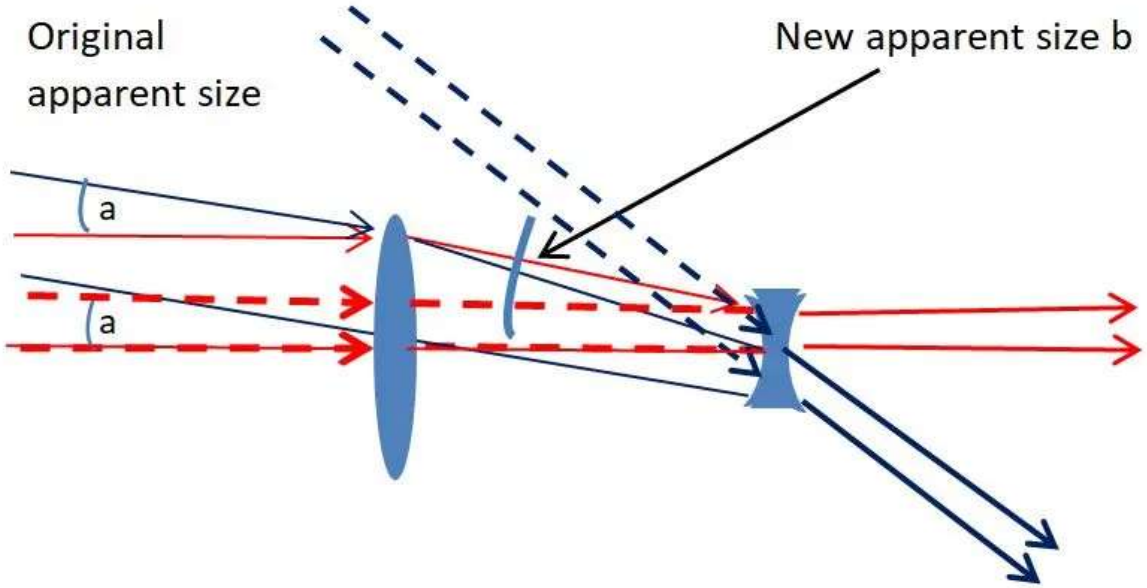
- Italian polymath: astronomer, physicist, engineer, philosopher, and mathematician
- Combined thought experiments (思想实验), mathematics, and observation
- “father of the scientific method”
- “father of observational astronomy”
- “father of modern physics”
- “father of science”

Galileo Galilei (1564-1642)



- Improved refracting telescope (折射望远镜) with a convergent (plano-convex 凸透镜) objective lens (物镜) and a divergent (plano-concave 凹透镜) eyepiece lens (目镜)





Galileo Galilei (1564-1642)

- **Celestial findings**

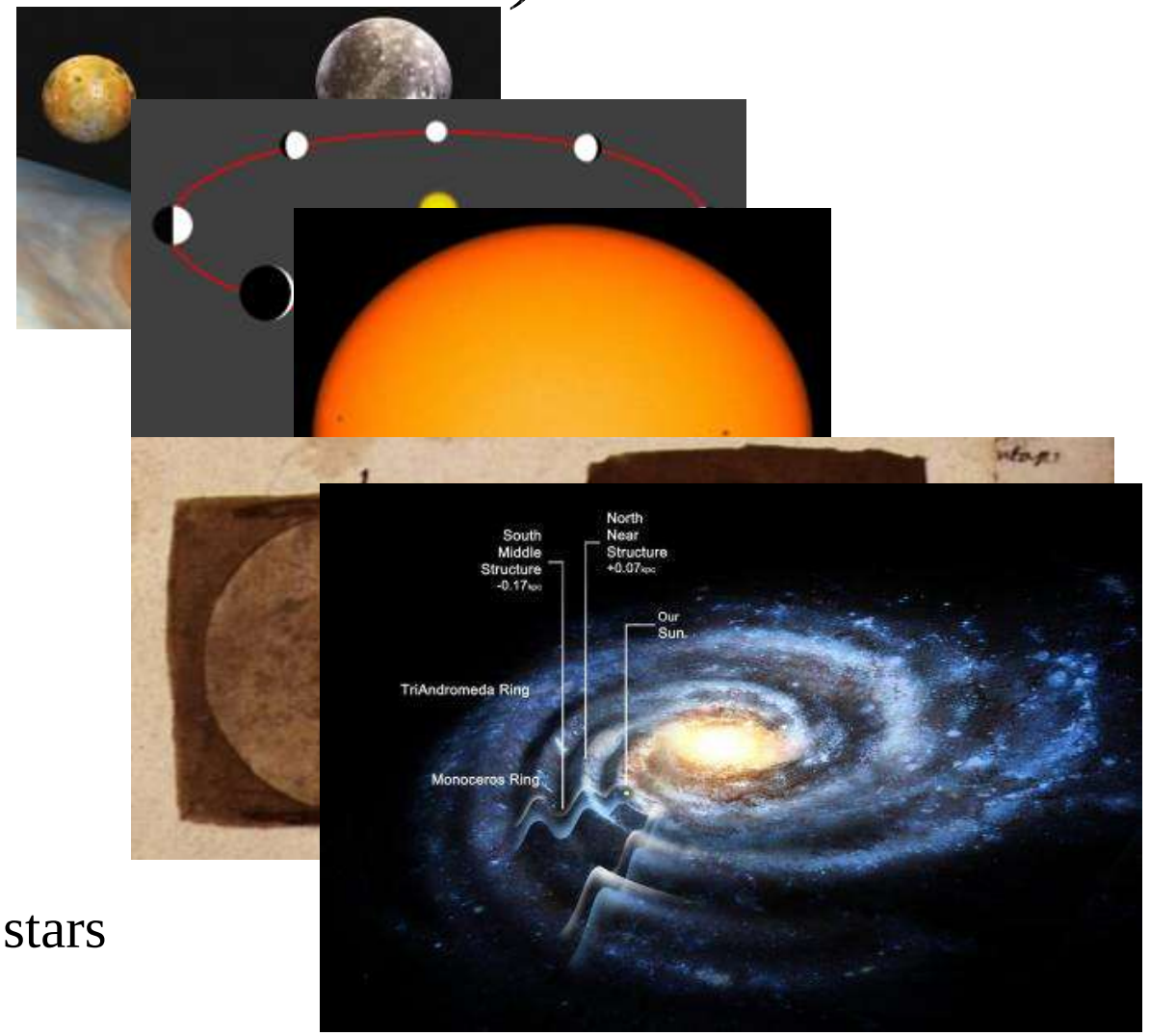
Satellites of Jupiter (木星)

A full set of phases of Venus (金星)

Sunspot (太阳黑子)

Topography (地形) of the moon

Milky way (银河) being a multitude of stars



The Trial (审判) of Galileo

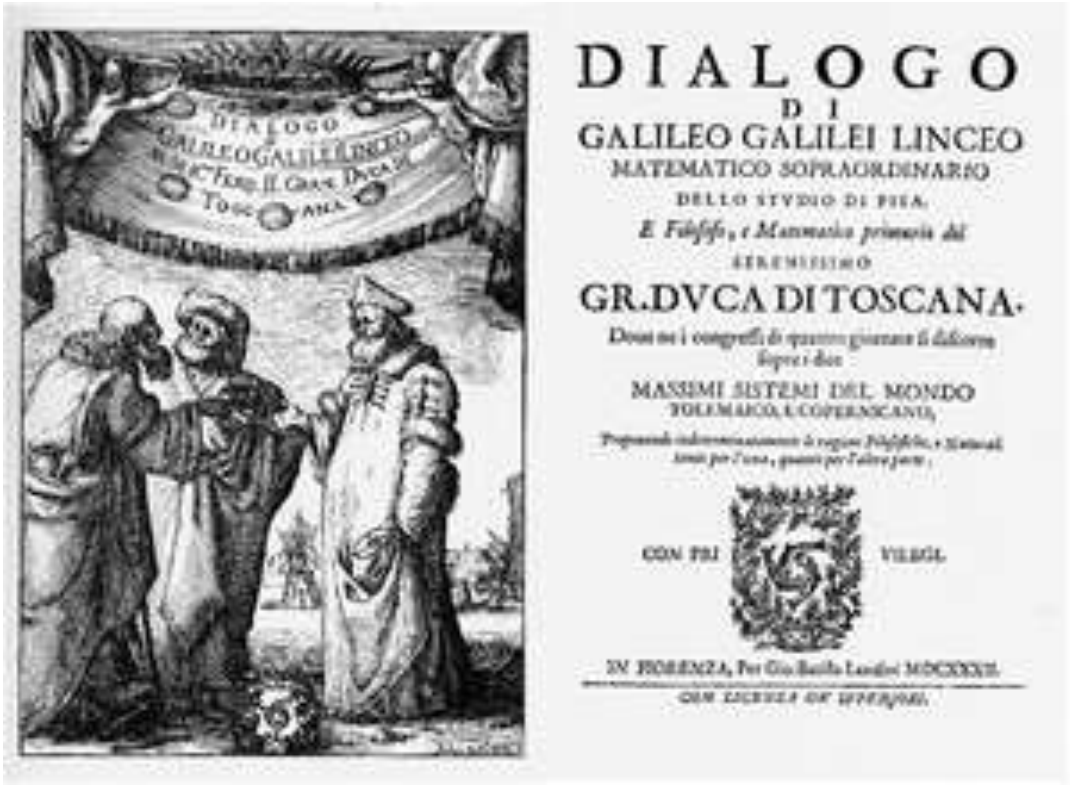
- March 1616

The Congregation of the Index banned Copernicus' *On the Revolutions* until corrections can be added. Galileo was censured (严厉谴责) by the Inquisition (宗教法庭).



Galileo Galilei (1564-1642)





DIALOGO
DI
GALILEO GALILEI LINCEO
MATEMATICO SOPRAORDINARIO
DELLO STVDIO DI PISA.
E Filosofo, e Matematico primario del
SERENISSIMO
GR.DVCA DI TOSCANA.
Doue ne i congressi di quattro giornate si discorre
sopra i due
MASSIMI SISTEMI DEL MONDO
TOLEMAICO, E COPERNICANO,
Proponendo indifferentemente, e con ragioni Filosofiche, e Naturali
tanto per l'vna, quanto per l'altra parte.
CON PRI VILEGI.
IN FIORENZA, Per Gio. Batista Landini MDCXXXII.
CON LICENZA DE' SVPERIORI.

Dialogue Concerning the Two Chief World Systems (1632)
《关于托勒密和哥白尼两大世界体系的对话》

- Galileo demonstrates the truth of the Copernican system over the Ptolemaic one.

The Trial (审判) of Galileo

- March 1616

The Congregation of the Index banned Copernicus' *On the Revolutions* until corrections can be added. Galileo was censured (严厉谴责) by the Inquisition (宗教法庭).

- April 1633

Galileo was interrogated (审问) before the Inquisition. Galileo was condemned as a heretic (异教徒) and his *Dialogue on the Two Chief World Systems* was banned.



Galileo's Science



- At the start of Scientific Revolution: science not guided by religion nor philosophy, not medieval approach
- Founded on his own actual measurements (approximate not exact)
- The world is a machine → analyzed by math
- Mathematics as the language for reading the "great book of nature"

"Nature is written in the language of mathematics, and its characters are triangles, circles, and other geometric figures...."

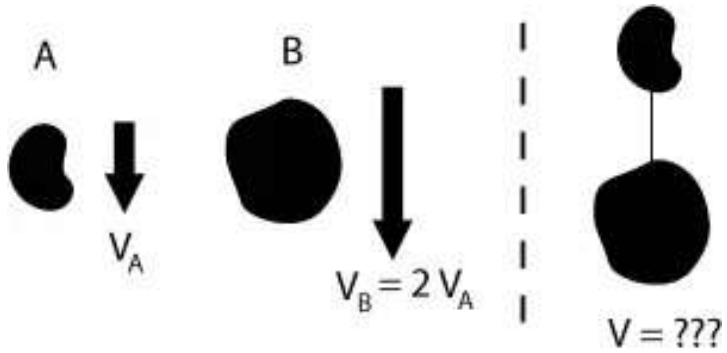
Galileo: leaning tower of Pisa experiment



Apollo 15: Physics on the Moon (1971)
On the moon: all objects released together fall at the same rate regardless of mass.

Galileo Galilei(1564-1642)

- **Thought experiment**



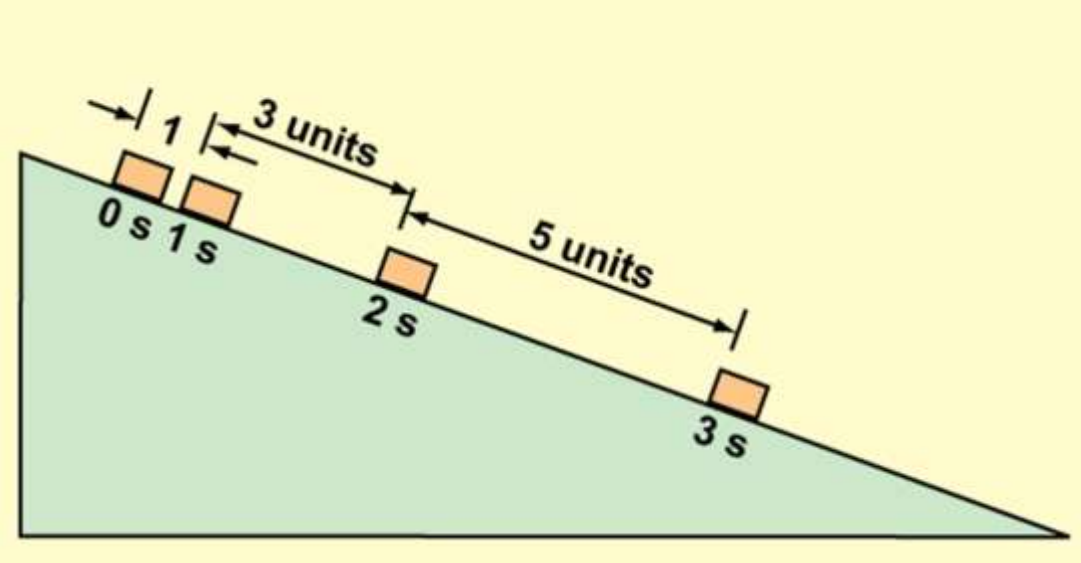
According to Aristotle, the heavier body will fall considerably faster than the light body (1), then if $m_B=2m_A$, $V_B = 2 V_A$(2).

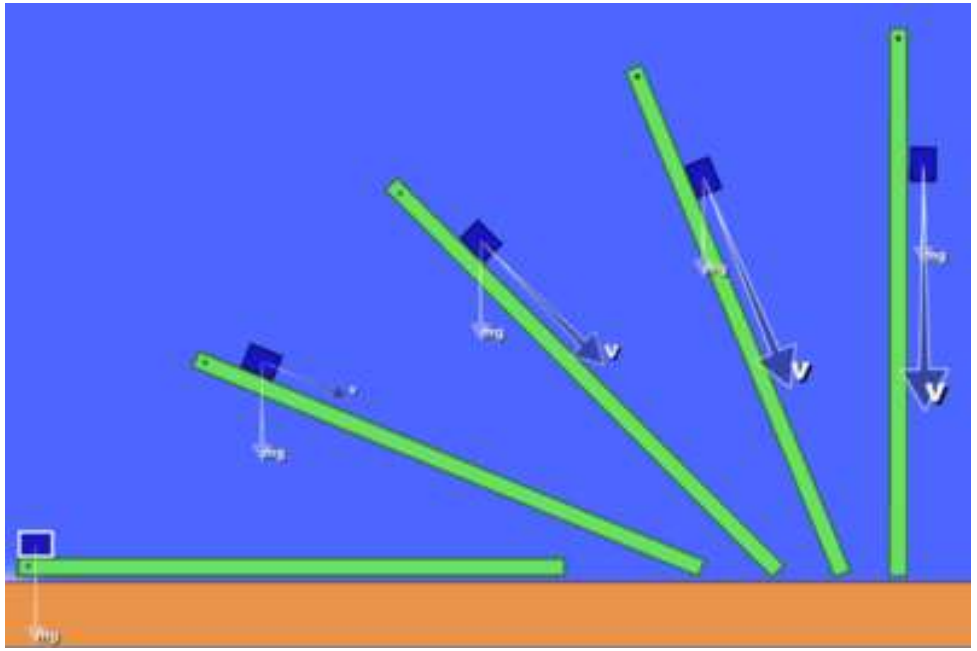
But what happens if the A and B are attached by a thread or a stick (3)?

Looked at in this way, Aristotle's ideas about falling bodies start to look absurd. (1) (2) (3)

Galileo Slope Experiment 1

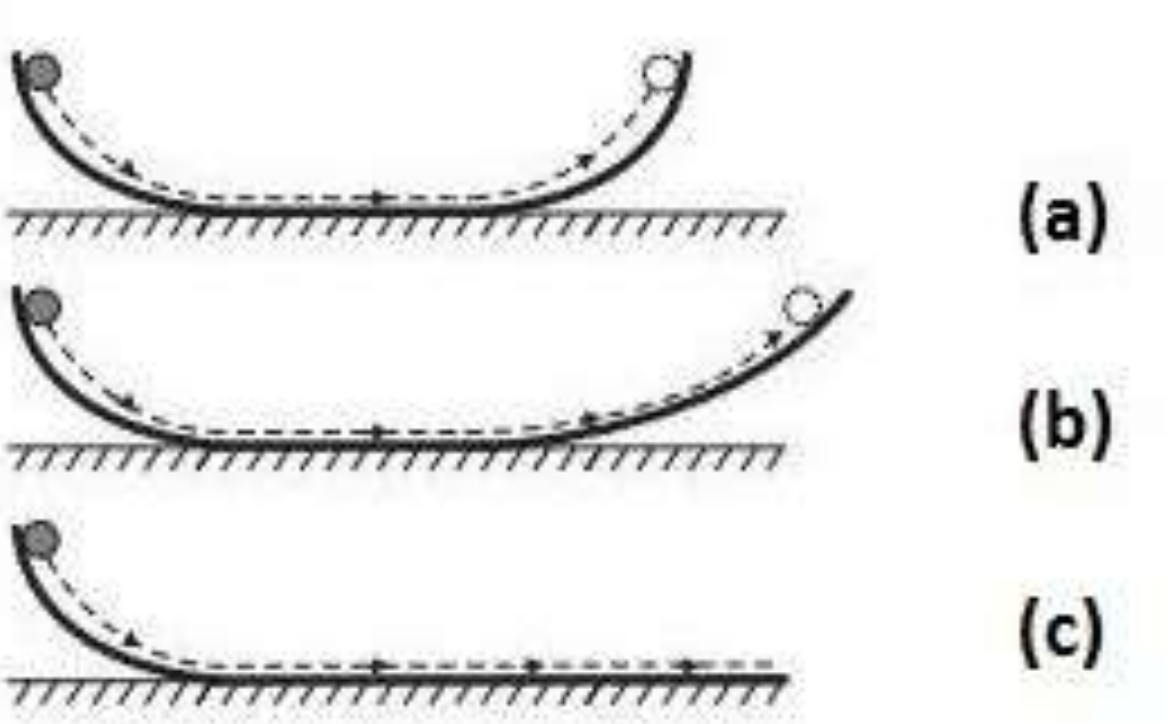
Falling is too fast. He let an object slide on a smooth inclined plane and discovered: no matter how the inclination angle changes, **velocity ∝ time** elapsed, **independent of mass.**





Uniformly Accelerated Motions

Galileo Slope Experiment 2



Galileo concluded that an object moving on a frictionless horizontal plane must neither have acceleration nor retardation, i.e. it should move with constant velocity.

Nicolas Copernicus (1473-1543)

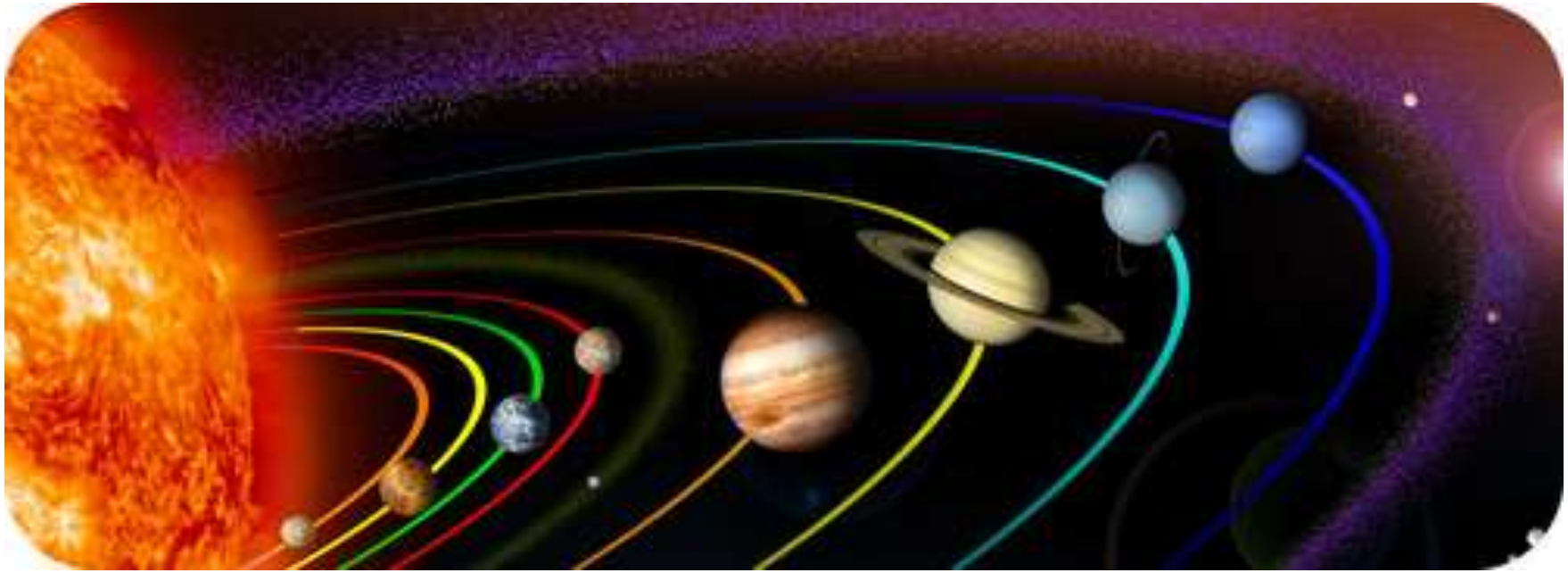
Copernicus's Mistake:

- *The Sun near the center of the Universe, motionless, with Earth and the other planets rotating around it in circular paths modified by epicycles and at uniform speeds.*

- *The planets orbit the sun circularly.*

Johannes Kepler (1571-1630)

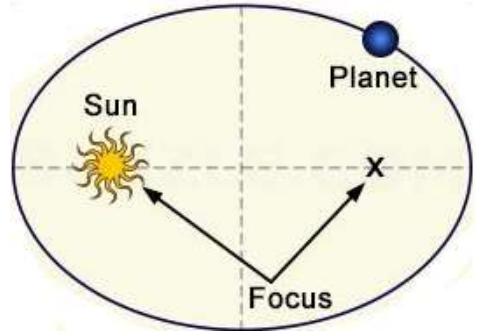
- German mathematician, astronomer, and astrologer.
- An assistant to the astronomer Tycho Brahe (第谷·布拉赫) in Prague.
- Famous for his planetary laws of motion.



Kepler's Laws (开普勒定律) of Planetary Motion

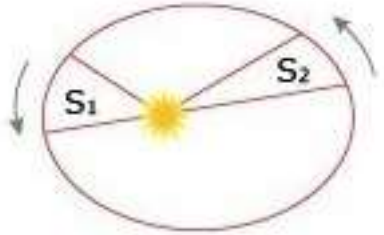
First Law: The orbit of every planet is an ellipse with the Sun at one of the two foci.(每一个行星都沿各自的椭圆轨道环绕太阳, 而太阳则处在椭圆的一个焦点中)

- Against Aristotle's claim that the motion is circular and the earth is the center



Second Law: A line joining a planet and the Sun sweeps out equal areas during equal intervals of time.(在相等时间内, 太阳和运动着的行星的连线所扫过的面积都是相等的)

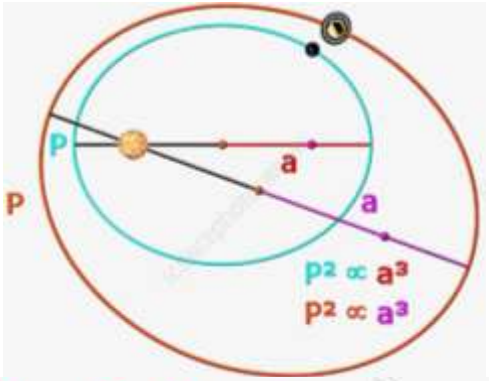
- Against Aristotle's claim that the motion is uniform, the second law requires that planetary motion changes



Equal area in the same time
area S1 = area S2

Third Law: The square of the orbital period of a planet is directly proportional to the cube of the semi-major axis of its orbit.(各个行星绕太阳公转周期的平方和它们的椭圆轨道的半长轴的立方成正比)

- Laid solid foundation for Newton's Laws of Motion



$P^2 \div a^3 = P^2 \div a^3 = \text{constant}$

Starter



Time: January of 1684

Venue: A coffee house in London

People: Robert Hooke

Edmond Halley

Sir Christopher Wren(雷恩爵士)

Topic: Whether the assumption of an inverse square law of gravity would lead to Keplerian(开普勒的) elliptical(椭圆的) orbits?

STARTER



-*“I’m offering a prize of a valuable book-token(书券) if either of you could provide a mathematical proof that this would be the case.”*



-*“I could but would not reveal my solution until you two have time to try the problem for yourselves.”*

-*“...I’m going to visit Newton.”*



PROCESS

August of 1684
Venue: Cambridge
People: Edmond Halley & Sir Isaac Newton


Halley

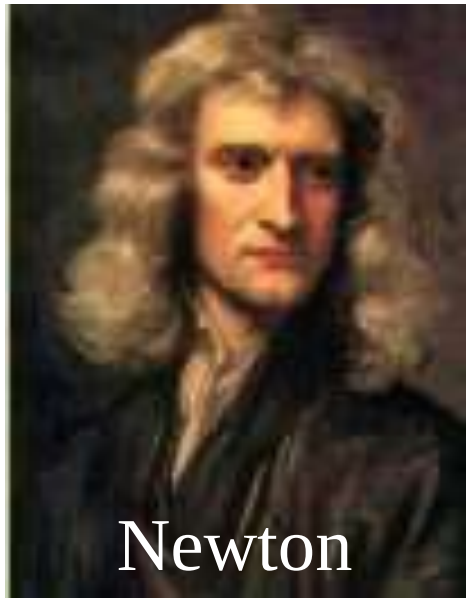
-“*Whether the assumption of an inverse square law of gravity would lead to Keplerian elliptical orbits?*”

-*”Under the assumption of an inverse square law of gravity the planetary orbits would be ellipses.”*

-*“how you could be so certain?”*

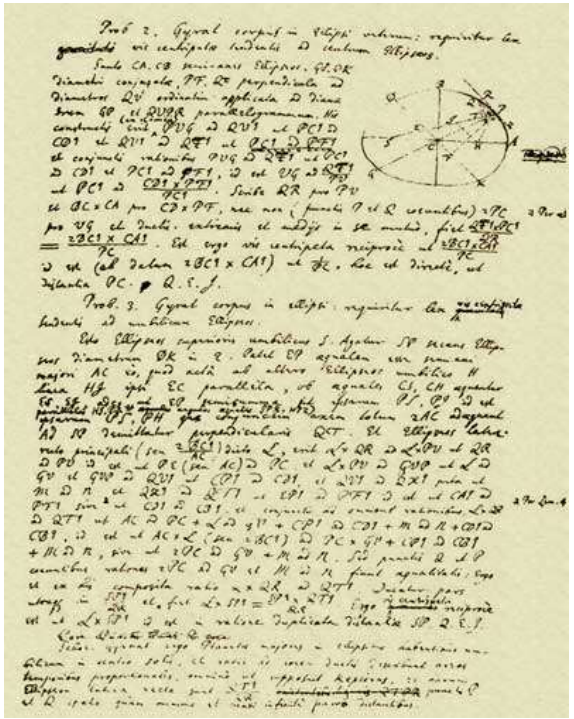
-*“because I have calculated it”*

-*“I have misplaced my calculations but I promise you that I would supply you with the necessary proof.”*


Newton

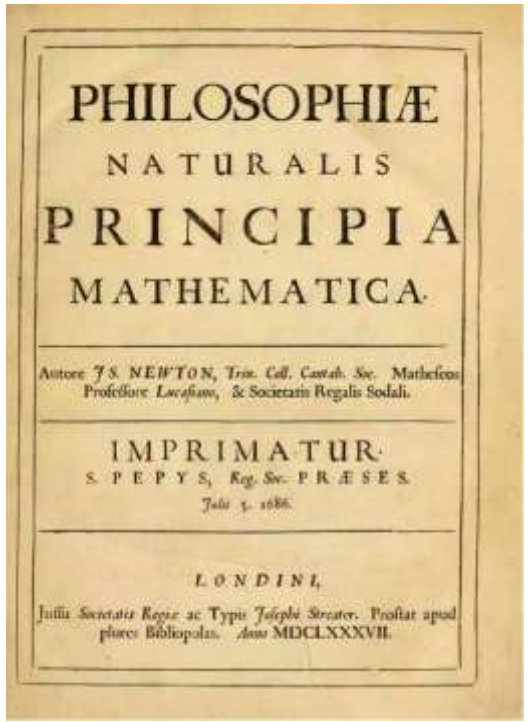
Consequence

November 1684



On the motion of bodies in an orbit
《论物体的运动》

July 1687



Mathematical Principles of Natural Philosophy (Principia)
《自然哲学的数学原理》(《原理》)

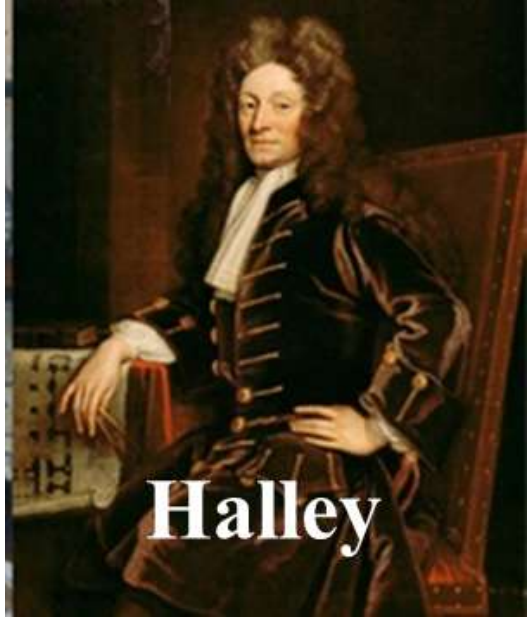
Principia

- Ode to Isaac Newton

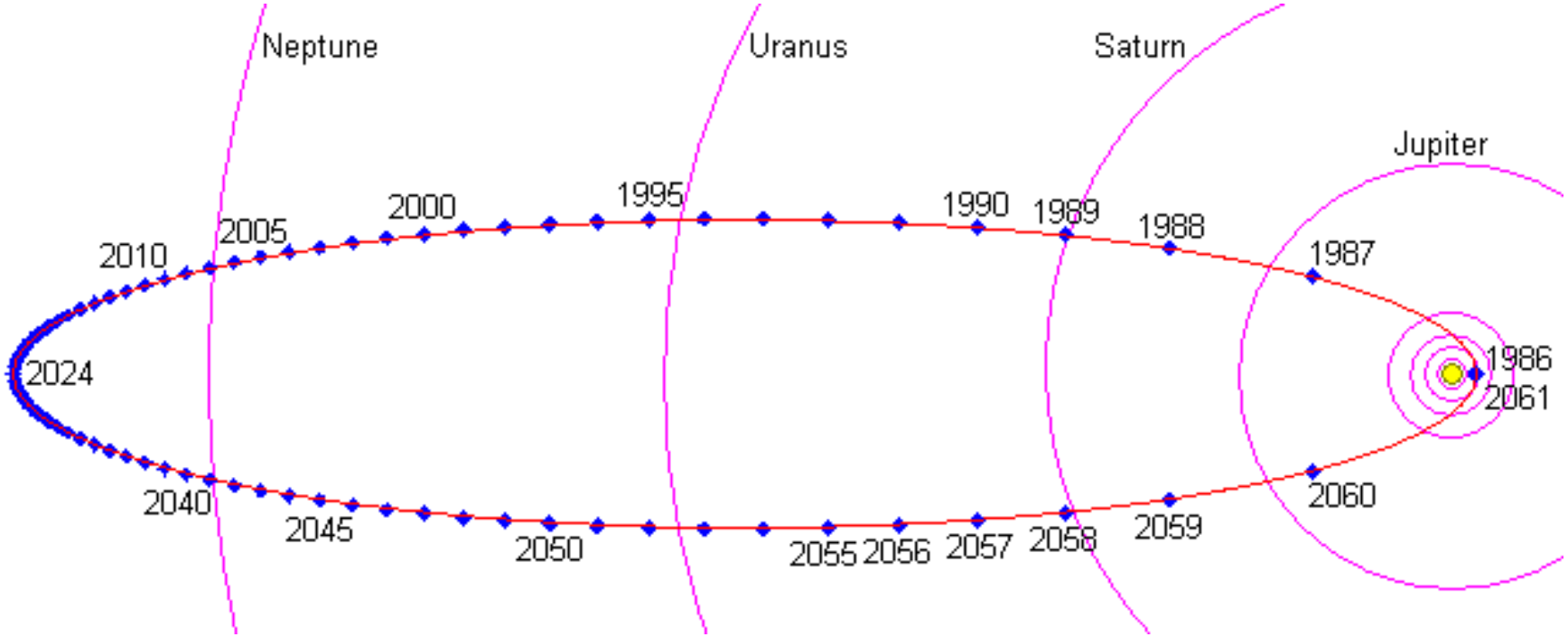
Then ye who now on heavenly nectar fare,
Come celebrate with me in song the name
Of Newton, to the Muses dear; for he
Unlocked the hidden treasuries of Truth:
So richly through his mind had Phoebus cast
The radiance of his own divinity.
Nearer the gods no mortal may approach.

你们，啊！饮天神美酒的人，和我一起歌颂牛顿的名字吧，他打开了隐秘真理的宝匣，牛顿，缪斯垂青的人，太阳神居住在他的心中，使他洋溢着神性，比任何一个凡人更接近神。

Edmond Halley(1656-1742)

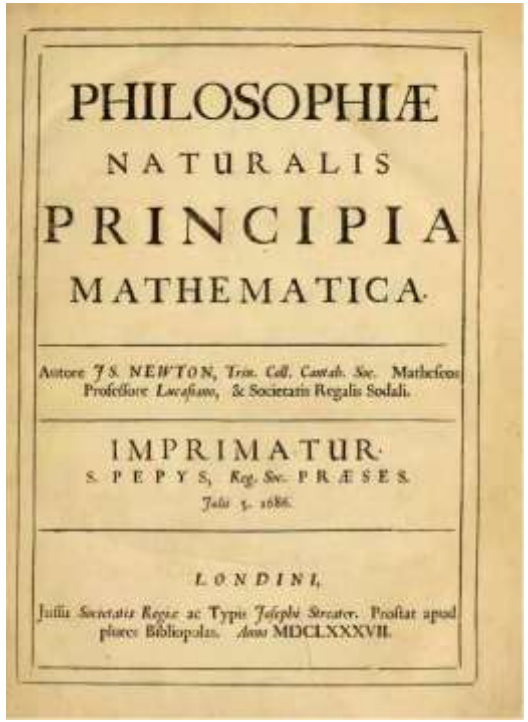


- English astronomer, geophysicist, mathematician, meteorologist, and physicist.
- Halley's comet (哈雷彗星 returns every 75.5 years. Last time: 1986. Next time: 2061.)



Contents of Principia

- Definitions (absolute space, time, and motion)
- Axioms (Laws of Motion)
- Book I: The Motion of Bodies
- Book II: Other Aspects in Physics
- Book III: The System of the World


PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA

Autore *Is. NEWTON, Trin. Coll. Cantab. Soc.* Matheseos Professore *Lucasiano*, & Societatis Regalis Sodali.

IMPRIMATUR
S. PEPYS, *Reg. Soc.* PRÆSES.
Julii 5. 1686.

LONDINI,
Juſſu *Societatis Regiæ* ac Typis *Joſephi Streater*. Proſtat apud plures Bibliopolas. *Anno* MDCLXXXVII.

What Did Newton Actually Do?

Making Definitions

- Statement that explains the meaning of the terms
- *E.g.* mass, motion, inertia, centripetal force.

Stating Axioms

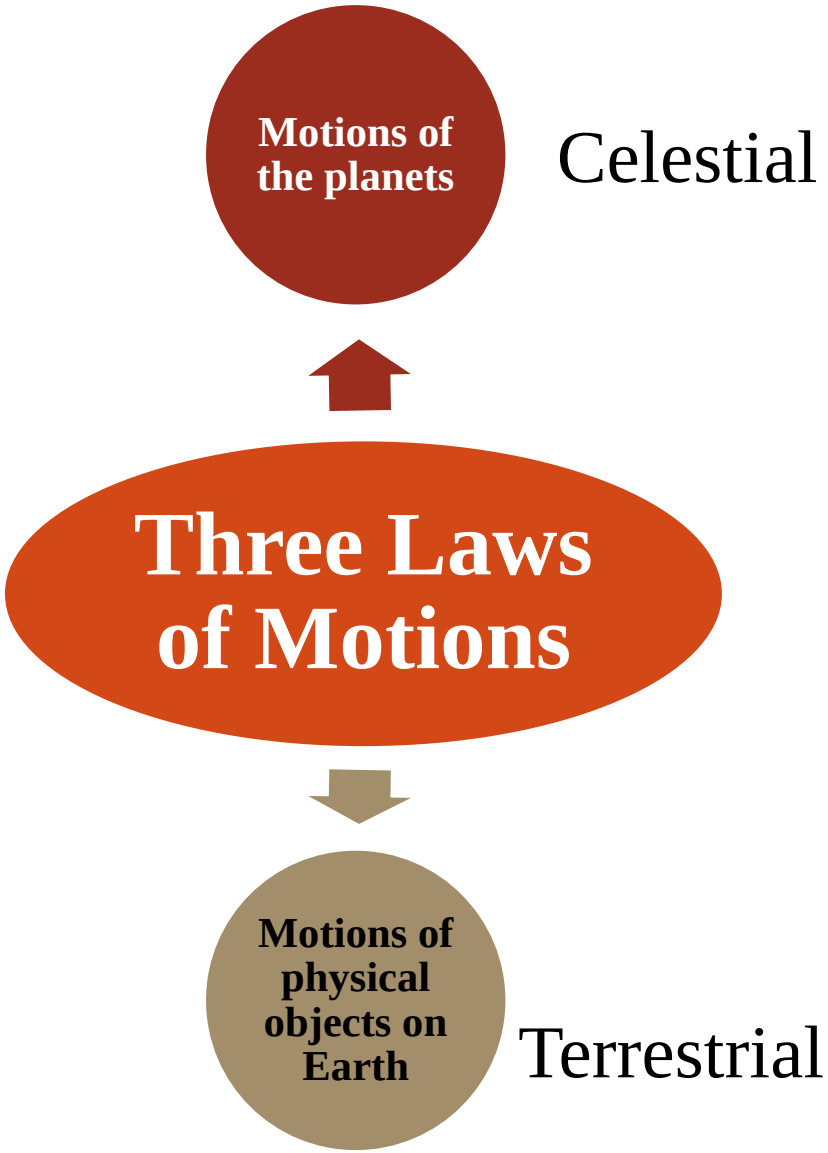
- Truth that cannot be proved but can only be accepted
- Newton's three laws of motion
- Newton's law of universal gravitation

Proving Results

- Projectile motion, planetary elliptic orbits



What Did Newton Actually Do?



Newton illustrated Law I by reference to projectiles that continue in their forward motions "so far as they are not retarded by the resistance of the air, or impelled downward by the force of gravity," and he referred also to "the greater bodies of planets and comets." (On the inertial aspect of the motion of "greater bodies" such as "planets and comets," see Supplement 12.) At this one stroke Newton postulated the opposite view of Aristotelian physics. In the latter, no celestial body could move uniformly in a straight line in the absence of a force, because this would be a "violent" motion and so contrary to its nature. Nor could a terrestrial object, as we have seen, move along its "natural" straight line without an external mover or an internal motive force. Newton, presenting a physics that applies simultaneously to both terrestrial and celestial objects, stated that in the absence of a force bodies do not necessarily stand still or come to rest as Aristotle supposed, but they may move at constant rectilinear speed. This "indifference" of all sorts of bodies to rest or uniform straight-line motion in the absence of a force

Comments from Cohen

- *"... the greatest achievement of Newtonian science must ever be the first full explanation of the universe on* ***mechanical principles*** *– one set of axioms and a law of universal gravitation that apply to all matter everywhere:* ***on earth as in the heavens*** *..."*

- **A new worldview!**

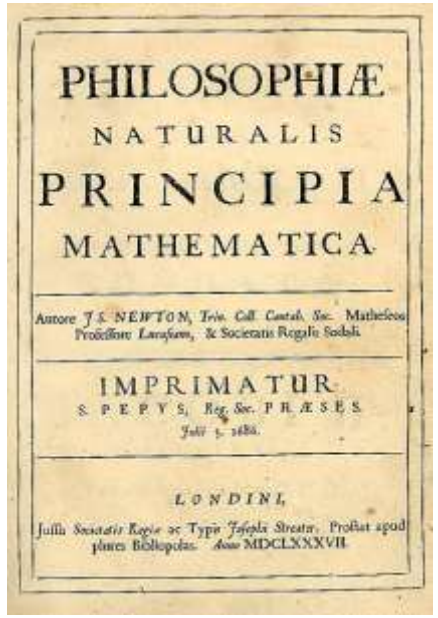
Isaac Newton (1643-1727)



Isaac Newton
1643-1727

"Nature and nature's laws lay hid in night; God said
"Let Newton be" and all was light."
--Alexander Pope, epitaph for Newton

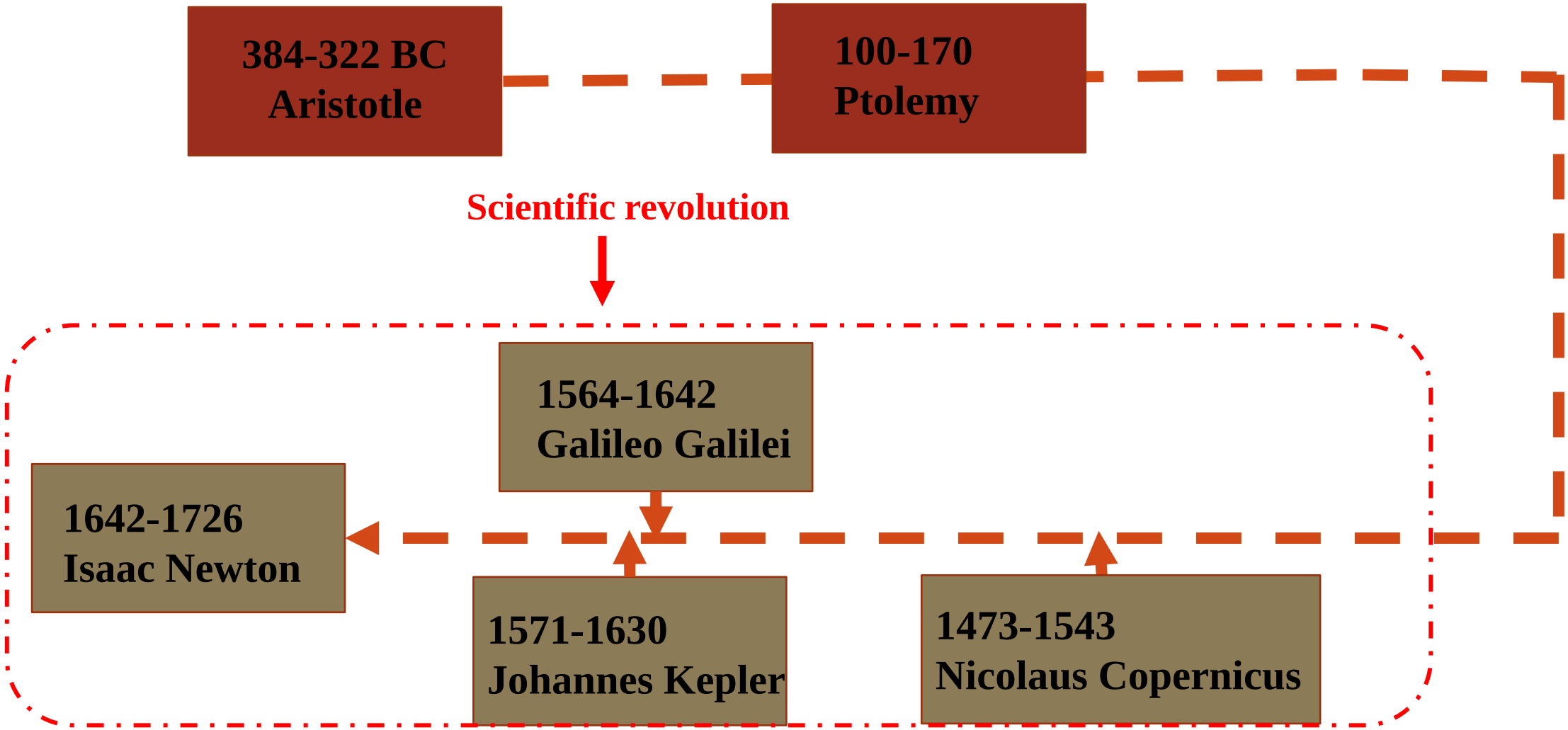
"Considered by many to be the greatest and most influential scientist who ever lived." (Wikipedia)


PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA

Autore J.S. NEWTON, Trin. Coll. Cantab. Soc. Matheseos Professore Lucasiano, & Societatis Regalis Sodali.

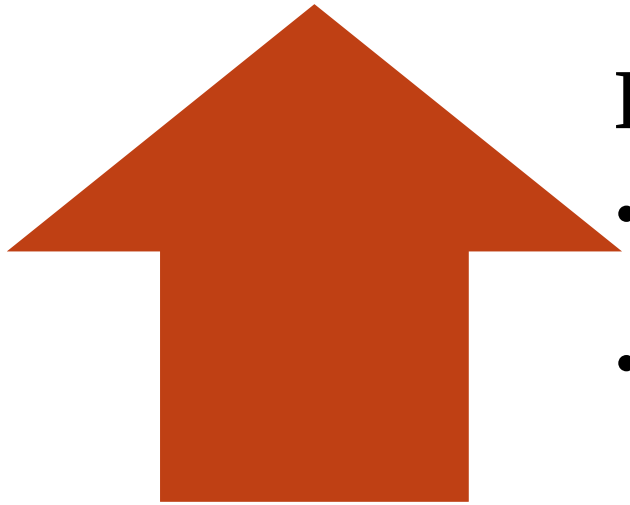
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S. PEPYS, Reg. Soc. PRÆSES.
Julii 5. 1686.

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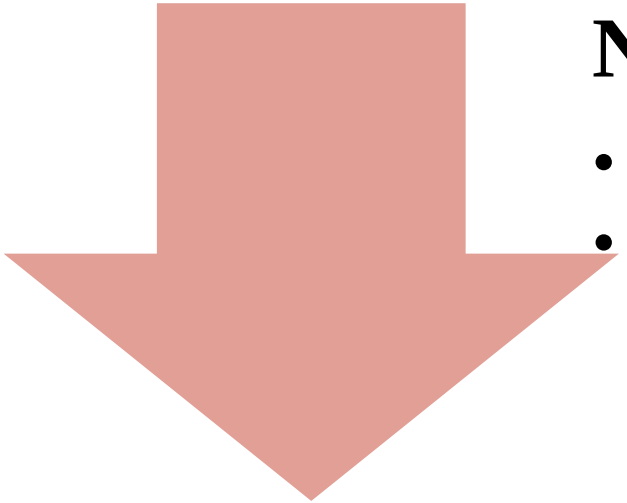
Science is a collective and a cumulative activity.

Two Different Senses of "Science"



Broader sense of science

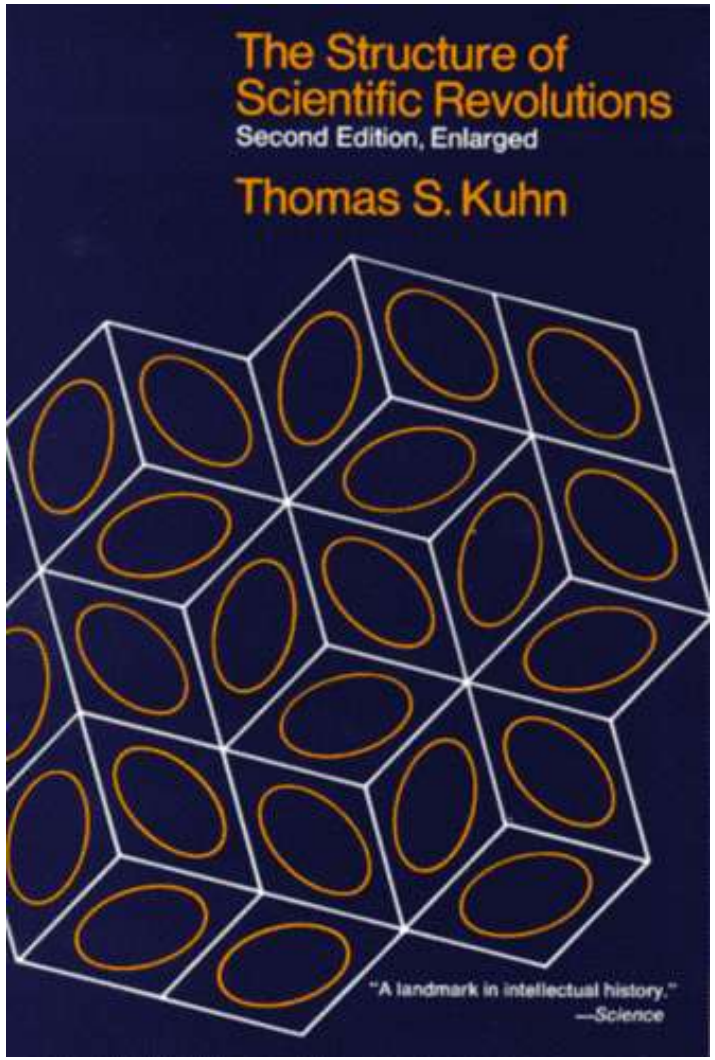
- The way of ***problem-solving*** based on ***observational*** evidence.
- Found in all human cultures



Narrower sense of science

- Emerged from Europe since 16th century
- The product after the Scientific Revolution (Copernicus, Galileo, Newton)

Paradigm Shift (范式转移)



- a concept identified by the American physicist and philosopher Thomas Kuhn (托马斯·库恩) in his influential book *The Structure of Scientific Revolutions(《科学革命的结构》)* in 1962

- a fundamental change in the basic concepts and experimental practices of a scientific discipline

Paradigm Shift (范式转移)

Examples [edit]

Natural sciences [edit]

Some of the "classical cases" of Kuhnian paradigm shifts in science are:

- 1543 – The transition in cosmology from a Ptolemaic cosmology to a Copernican one.[12]
- 1543 – The acceptance of the work of Andreas Vesalius, whose work *De humani corporis fabrica* corrected the numerous errors in the previously-held system created by Galen.[13]
- 1687 – The transition in mechanics from Aristotelian mechanics to classical mechanics.[14]
- 1783 – The acceptance of Lavoisier's theory of chemical reactions and combustion in place of phlogiston theory, known as the chemical revolution.[15][16]
- The transition in optics from geometrical optics to physical optics with Augustin-Jean Fresnel's wave theory.[17]
- 1826 – The discovery of hyperbolic geometry.[18]
- 1859 – The revolution in evolution from goal-directed change to Charles Darwin's natural selection.[19]
- 1880 - The germ theory of disease began overtaking Galen's miasma theory.
- 1905 – The development of quantum mechanics, which replaced classical mechanics at microscopic scales.[20]
- 1887 to 1905 – The transition from the luminiferous aether present in space to electromagnetic radiation in spacetime.[21]
- 1919 – The transition between the worldview of Newtonian gravity and general relativity.
- 1964 - The discovery of cosmic microwave background radiation leads to the big bang theory being accepted over the steady state theory in cosmology.
- 1965 - The acceptance of plate tectonics as the explanation for large-scale geologic changes.
- 1974 - The November Revolution, with the discovery of the J/psi meson, and the acceptance of the existence of quarks and the Standard Model of particle physics.
- 1960 to 1985 - The acceptance of the ubiquity of nonlinear dynamical systems as promoted by chaos theory, instead of a laplacian world-view of deterministic predictability.[22]



Content of the Course

Physical Universe

- Plato, Lindberg
- Aristotle, Lindberg
- Cohen, Newton

World of Life

- Darwin
- Watson
- Carson

Human Understanding

- Poincare
- Kandel
- Needham
- Sivin, Shen Kuo
- Dunhan, Euclid

RJ topics

Topic 1	Part I introduces the origin of modern science. How the perspectives on the natural world held by Plato, Aristotle, and Newton influenced the evolution of modern science?
Topic 2	Part I of our textbook, *In Dialogue with Nature*, explores scientific developments in early times, beginning with Plato's Allegory of the Cave, whereas many other History of Science textbooks start with Aristotle. What could be the reason for this arrangement in our textbook? Please explain the reason based on the content of Part I.



RJ Requirements

Remarks:

a. Please refer to the grading guideline and template on Blackboard to write your paper.
b. Word limit: 600-800 for English, 900-1200 for Chinese.
c. Submit your assignment on VeriGuide. Late submission is not allowed.
d. The filename should be in this format: RJ-YourName.doc/docx (e.g., RJ-ChanTaiMan.doc/docx).
e. Deadline: Mar 9 (Sun, 23:55), 2025.

Reading Assignment

- Read Text 3.